

Mohit Manna

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EDUCATION

Purdue University

Dec 2027

BS Computer Engineering & Math

Minor: Finance

- **Orgs/Awards:** 1st/200: Purdue Engineering Research Symposium · ML @ Purdue · Purdue Stack

EXPERIENCE

IBM | *Software Engineer Intern*

May – Aug 2026

- Built Go tooling used by **42** engineers across **6** OpenShift namespaces, cutting manual setup time by **90%**
- Designed a Linux subprocess scheduler for **30** dependency-ordered setup stages with bounded **4**-way concurrency
- Hardened **15** Bash workflows with retries and timeouts, eliminating **~5** manual retry steps per setup

Caterpillar | *Machine Learning Intern*

Aug – Apr 2026

- Developed a PyTorch multi-horizon forecasting model on **1M+** supply-chain records, improving accuracy by **12%**
- Scaled batched inference across **520** SKUs on Docker/Kubernetes, cutting runtime to **2.85** min
- Automated retraining with **4**-week shadow validation and metric gates before rollout to **20** Angular dashboards

Creative Capital | *Software Engineer Intern*

Jun – Aug 2025

- Shipped React market dashboard with GraphQL, supporting **~500** live connections and **10K+** records
- Built AI property Q&A across **23** tools with Pydantic-validated outputs, serving **10K+** users
- Secured **40+** Express endpoints with JWT auth and Postgres RLS, validated by integration tests

Photonic Simulation Lab | *Computational Photonics Research Assistant*

Jan – Apr 2025

- Optimized C++ simulation kernels and Python scheduling, cutting runtime by **15%** at **<75** ms p95 dispatch latency
- Parallelized **7K** waveguide parameter sweeps across **70** workers, validating loss via mesh-convergence tests
- Implemented numerical waveguide models to extract effective index/propagation loss across geometry sweeps

PROJECTS

AegisFeed – *High-Performance Market Data Engine in C++ 20*

- Benchmarked C++20 Nasdaq ITCH parsing + book reconstruction at **~5M** msgs/sec
- Stress-tested a bounded **65K**-message reorder buffer under injected UDP loss and reordering
- Recovered **1.2M** messages under simulated UDP loss with **0** state mismatches against clean replay

PulseKV — *Low-Latency Key-Value Engine in C*

- Benchmarked an epoll-based C key-value server at **~192K** reads/sec across **500** TCP clients
- Implemented async WAL group commits and batched crash recovery, cutting **20K**-record replay time by **~80%**

Sum100 — *Prediction Market Coherence Engine in Rust (Kalshi)*

- Built real-time Rust order-book engine across **80+** Kalshi contracts with **4** fee-aware arbitrage checks
- Verified deterministic replay over **14K+** events with zero book-state mismatches and fill-or-kill execution gates

SKILLS

- **Languages:** C++ · C · Python · Rust · Go · Java · SQL · TypeScript
- **Frameworks:** PyTorch · FastAPI · Pydantic · Node.js · Express · React · GraphQL
- **Tools:** Linux · Git · Docker · Kubernetes · PostgreSQL · Redis · AWS · Azure · OpenTelemetry